

CO1 Assessment Tool 2

Constraint-Based Problem Solving

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Course Code:- CSA1709 - Artificial Intelligence

1. Doctor–Shift Assignment using Backtracking

Doctors: D1, D2, D3

Shifts: Morning (M), Afternoon (A), Night (N)

Constraints

- $D1 \neq N$
- D2 must be before D3
- One doctor per shift
- $D3 \neq M$
- Every doctor gets one shift

Backtracking

Start with **D1**:

- $D1 = M$ ✓
- $D2 = A$ ✓
- $D3 = N$ ✓
- D2 before D3: $A < N$ ✓

So a valid solution is found.

Final Schedule:

Shift	Doctor
Mornin g	D1
Afterno on	D2
Night	D3

Constraint check:

D1 $\neq$ N ✓ | D2 < D3 ✓ | One/shift ✓ | D3 $\neq$ M ✓ | All assigned ✓

Final: D1–Morning, D2–Afternoon, D3–Night

2. Grid Robot – BFS with Manhattan Heuristic

Important: The actual 5×5 grid/obstacle positions are not provided in the question text. Therefore, the exact expansion order and path cannot be determined without the grid.

Given

- Movement: Up, Down, Left, Right
- Move cost = 1
- Obstacles cannot be crossed
- Manhattan heuristic:

$$h(n) = |x_n - x_G| + |y_n - y_G| \quad h(n) = |x_n - x_G| + |y_n - y_G|$$

BFS Procedure

1. Put **S** into the queue.
2. Remove the first node from the queue.
3. Expand its valid neighbours.
4. Ignore obstacles and already visited cells.
5. Add valid neighbours to the queue.
6. Continue until **G** is reached.

Priority/Queue Example

Queue: [S]



Expand S



Add valid neighbours



Queue: [A, B]



Expand A



Add new neighbours



Continue until G

Since every move costs **1**, BFS guarantees the **shortest path**.

Optimal cost = number of moves in the shortest path.

The exact node expansion, queue updates and final path require the missing 5×5 grid.

3. Rescue Robot – Online Search

a) Effect of Constraints

- **4-direction movement:** Limits possible actions.
 - **Obstacles:** Cannot enter blocked cells.
 - **Limited sensing:** Robot does not know the complete environment.
 - **Movement cost = 1:** Normal movement has low cost.
 - **Risky zone = +2:** Robot should avoid unnecessary risky areas.
 - **Dynamic knowledge:** Robot must update its map while moving.
 - **Goal:** Reach survivor with minimum cost and safely.
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b) Solution Approach

Online Search + Uniform Cost Search

The robot initially knows only its nearby surroundings.

1. Start at **S**.
2. Sense adjacent cells.
3. Add discovered cells to its internal map.
4. Assign cost:
 - Normal cell = **1**
 - Risky cell = **3** (1 + 2)
5. Select the lowest-cost path.
6. Move to the next cell.
7. Sense again and update the map.
8. Recalculate the path if a new obstacle/risk is discovered.
9. Continue until **G** is reached.

UCS chooses the path with minimum total cost.

c) AI Concepts

Intelligent Agent

The robot **perceives** the environment using sensors and **acts** by choosing movements.

Rationality

It chooses the action that minimizes **cost and risk** while trying to reach the survivor.

Environment Type

Property	Environment
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Observability	Partially observable
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Determinism	Stochastic/unertain
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State	Dynamic
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Agents	Mainly single-agent
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Actions	Discrete
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Knowledge	Initially incomplete
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d) Why This Approach?

Online Search + UCS is suitable because:

- The robot has **limited information**.
- It discovers obstacles while moving.
- UCS finds the **minimum-cost path**.
- It considers extra costs of risky zones.
- The robot can **re-plan dynamically** when new information is discovered.

Final Summary

Sense → Update Map → Calculate Costs → Choose Minimum-Cost Path → Move → Sense Again → Re-plan → Reach G